

C. Both outliers and spurious correlation

Solution:

C is correct. Both outliers and spurious correlation are potential problems with interpreting correlation coefficients.

Use the information in Exhibit 29 to answer questions 3 and 4.

An analyst is evaluating the tendency of returns on the portfolio of stocks she manages to move along with bond and real estate indexes. She gathered monthly data on returns and the indexes:

Exhibit 29: Monthly Data on Returns and Indexes

	Returns (%)		
	Portfolio Returns	Bond Index Returns	Real Estate Index Returns
Arithmetic average	5.5	3.2	7.8
Standard deviation	8.2	3.4	10.3
	Portfolio Returns and Bond Index Returns	Portfolio Returns and Real Estate Index Returns	
Covariance	18.9	-55.9	

3. Without calculating the correlation coefficient, the correlation of the portfolio returns and the bond index returns is *most likely* to be:

- A.** negative.
- B.** zero.
- C.** positive.

Solution:

C is correct. The correlation coefficient is positive because the covariance is positive.

4. Without calculating the correlation coefficient, the correlation of the portfolio returns and the real estate index returns is:

- A.** negative.
- B.** zero.
- C.** positive.

Solution:

A is correct. The correlation coefficient is negative because the covariance is negative.

5. Consider two variables, *A* and *B*. If variable *A* has a mean of -0.56 , variable *B* has a mean of 0.23 , and the covariance between the two variables is positive, the correlation between these two variables is:

- A.** negative.
- B.** zero.